DAY 4 | PRE-READ

How Apps Talk and Remember

A light read before Day 4. About 15 minutes, no code required.

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Why read this?

When you raise a complaint in a telecom app, your phone, the company's servers and a database all work together in a second or two. Tomorrow you will see how that works, draw your first charts and store SmartDesk tickets in a modern database called MongoDB.

1. Front of house and back of house

The **frontend** is what you see: screens, buttons and forms. The **backend** is the part you never see: the rules, the calculations and the data. Between them, **middleware** checks every request on the way in, like a security desk in an office building.

2. The restaurant model of an app

Restaurant	App world	Job
Customer	Client (phone, browser)	Asks for something
Waiter	API	Carries requests and responses, follows the menu rules
Kitchen	Server	Does the actual work
Pantry	Database	Stores the ingredients, which is the data

The customer never walks into the kitchen. In the same way, apps never touch the database directly. They talk to an API that follows agreed rules.

3. Pictures make numbers speak

A table of 365 daily ticket counts is hard to read. A simple line chart shows the trend instantly. We will use a Python tool called **Matplotlib** to draw charts.

4. Two ways to store information

Some databases store data in neat tables, like a spreadsheet. Others store each record as a flexible **document**, like a filled-in form that can have extra sections when needed. MongoDB is a document database.

> **Example:** one customer has two phone numbers and another has five. A document simply holds a list of numbers, without needing a separate table.

5. Finding things fast

A book's index lets you jump to a topic without reading every page. A database **index** does the same for your data. It makes searches fast, at the cost of a little extra work every time you add data.

6. Four actions on any data: CRUD

- **Create:** log a new support ticket.
- **Read:** search for a customer's tickets.
- **Update:** change a ticket's status to closed.
- **Delete:** remove a test ticket.

7. Think about these before class

- Pick an app you used today. What was the client, and what data did it ask for?
- What information would you store on a support ticket? Is any of it a list?
- Which chart would you use to show tickets per priority?

Irasshaimase (いらっしゃいませ): welcome, said to customers in shops and restaurants